

Smart Stage

Smart Anchor & Stage Flow Management System

- **Plan the event**
- **Manage the flow**
- **Adapt in real time.**

TEAM NAME

Brainstormers

TEAM LEADER

Dhwani Shah

PROBLEM STATEMENT

PS-5 · Smart Anchor & Stage Flow Management System

COLLEGE / INSTITUTION

CHARUSAT - Charotar University of Science and Technology

Live events break when the schedule does



The problem

- Anchors juggle intros, timings and announcements at once
- Schedules slip, so prepared scripts and cues go stale
- Speaker details are scattered across chats and sheets



Who it is for

- Anchors and hosts on stage
- Organizers and event coordinators
- Speakers, judges and guests
- Volunteers and backstage teams



Why it matters

- Delays and dead air lose the audience
- Wrong names or titles hurt credibility
- Hackathons and fests run on tight slots

One live agenda, one AI co-anchor

KEY IDEA

AI-powered event management that keeps every stage running smoothly adapts instantly to every change

Dynamic Stage Assistance



Plan ahead

Events, agenda, speakers and guests in one place



Generate scripts

AI writes the opening, introductions, transitions and closing



Run live

A dashboard shows the current and upcoming activity



Adapt instantly

Delays and surprise announcements reflow the agenda and refresh scripts

TECHNOLOGY STACK & ARCHITECTURE

How the pieces fit together

LANGUAGES

JavaScript (Node.js), HTML, CSS

FRAMEWORKS & LIBRARIES

React · Tailwind CSS · Express · Socket.IO

DATABASE

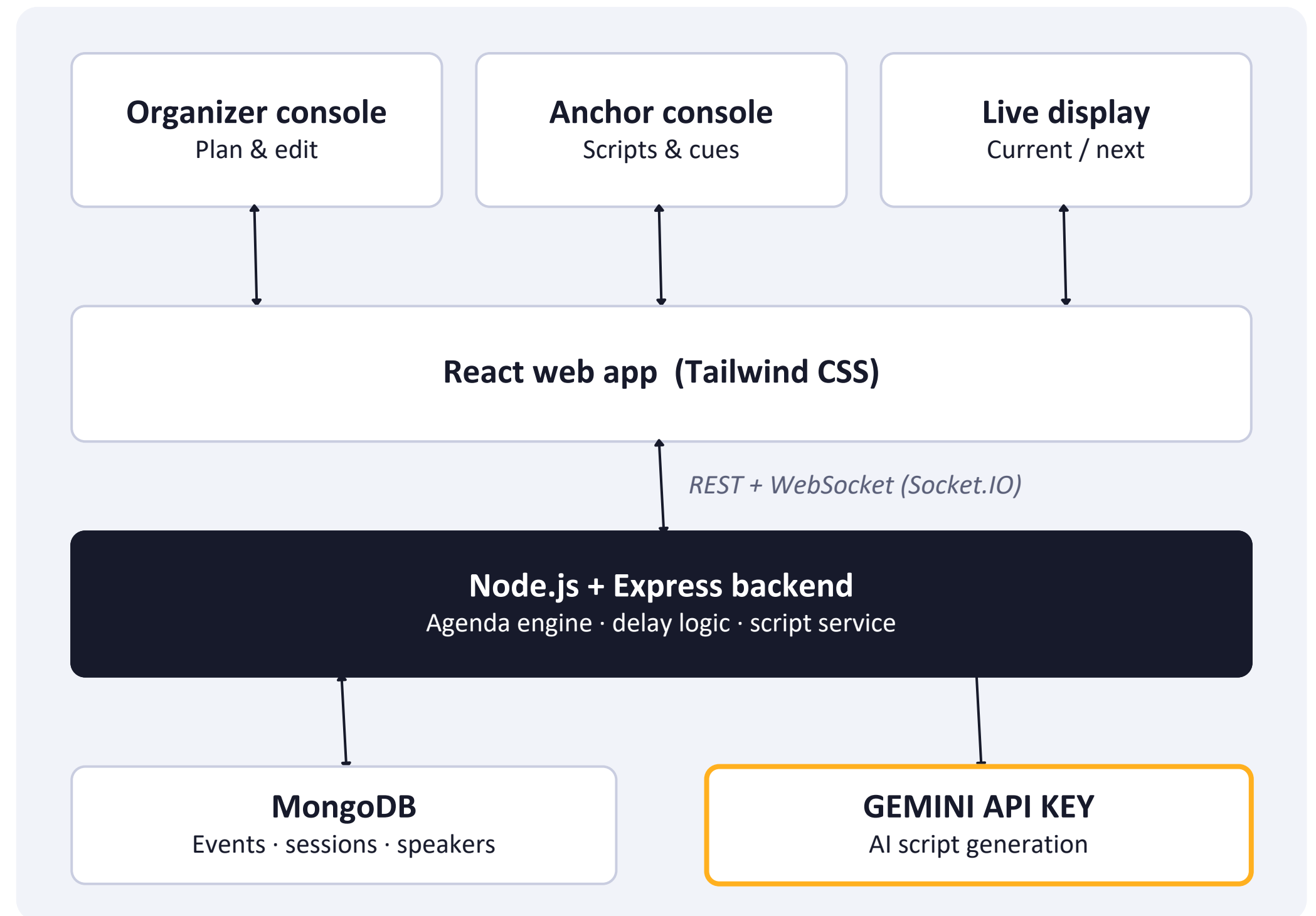
MongoDB (Atlas)

APIs / EXTERNAL SERVICES

Gemini API

DEVELOPMENT TOOLS

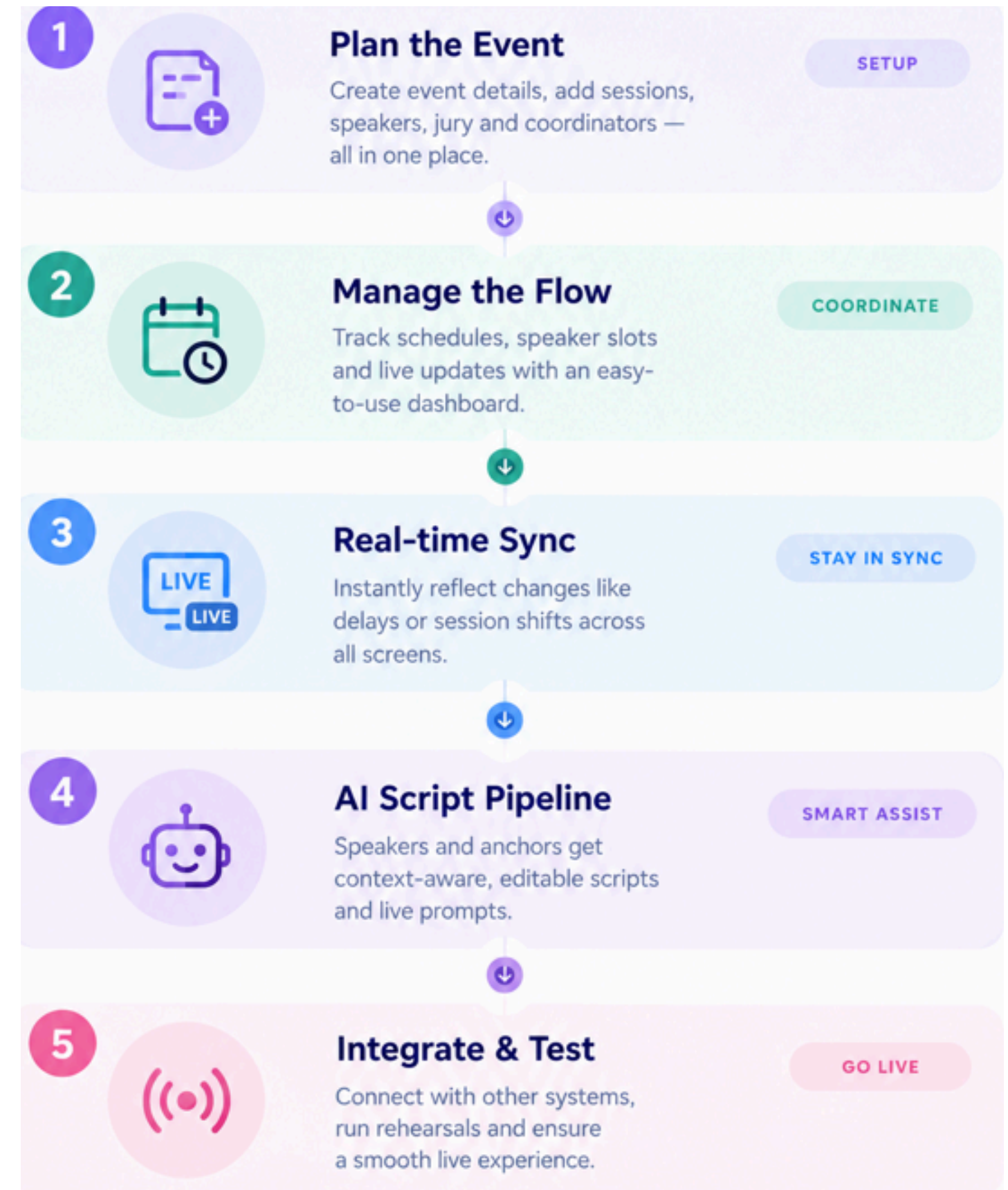
Git & GitHub · VS Code · Postman



From Planning to Live Execution

Key techniques

- **Context-aware prompting** : Uses event data to generate relevant and natural scripts
- **Delay & dynamic updates**: Automatically adjusts the flow and keeps everyone in sync when there are delays or changes.
- **Live Updates** : Inserts live updates (announcements, transitions, alerts) into the script in real time, then refresh the nearby transition



Core Features & Progress

IMPLEMENTED

- ✓ Event & agenda management
- ✓ Speaker & guest management
- ✓ AI opening script
- ✓ AI speaker introductions
- ✓ AI transition scripts
- ✓ AI closing script
- ✓ Live dashboard: current & upcoming
- ✓ Dynamic agenda updates
- ✓ Delay & announcement handling

Team Contributions



Member 1

Reha Shah

Backend & API

- REST + Socket.IO server
- Agenda & delay logic



Member 2

Rishva Prajapati

Frontend

- React dashboard
- Anchor & organizer views



Member 3

Astha Patel

AI & Database

- Prompt design & script flow
- MongoDB schema



Member 4

Dhwani Shah

UI/UX & Testing

- Design system & flows
- Scenario testing & demo

The product in action

SMART_STAGE

LIVE OPERATING PLATFORM

ACTIVE CONSOLE

Organizer

ORGANIZER TOOLS

Dashboard

Agenda Builder

Speakers & Team

AI Script Studio

Live Stage Control

BACKSTAGE OPS

Backstage Control

ANCHOR LECTERN

Stage Teleprompter

Event Brief

Joined as TechFest 2026

Landing / Switch Event

TechFest 2026

TF2026

Main Auditorium, Block A · 2026-02-14

Organizer

Coordinator

Anchor Stage

Live Sync

04:21:32 PM

TechFest 2026

READY

Code: TF2026

Main Auditorium, Block A · Date: 2026-02-14 · Organizer PIN: 1234

START LIVE EVENT

AI Script Studio

TOTAL AGENDA

5 Sessions

Planned sequence active

SPEAKERS & GUESTS

3 Assigned

Grounded speaker profiles

SCHEDULE DRIFT

+365 min

Live cascade delta

STAGE STATUS

Welcome Address

At 16:15

EVENT READINESS CHECKLIST

Agenda & Timing Defined

Keynote & Speakers Grounded

AI Script Studio Ready

ROLE ACCESS & LINKS

Team members can join instantly with the 6-character code TF2026.

LIVE STAGE FLOW

Sessions ordered by sequence with recomputed times

Edit Agenda

#	Session	Planned	Computed	Speaker / Room	Status
1	Opening Ceremony 10 min session	10:00	16:00	— Main Auditorium	COMPLETED
2	Welcome Address 10 min session	10:10	10:10 16:15	Dr. Rao Main Auditorium	DELAYED
3	Keynote — Future of AI 40 min session	10:20	10:20 12:42	Dr. Mehta Main Auditorium	COMPLETED
4	Workshop — Building with LLMs 60 min session	11:00	11:00 12:57	Ms. Patel Room C	COMPLETED
5	Refreshment Break 20 min session	12:00	15:53	— Foyer	COMPLETED

Live dashboard: current & upcoming

SMART_STAGE
LIVE OPERATING PLATFORM

ACTIVE CONSOLE
Anchor TF2026

ORGANIZER TOOLS
Dashboard >
Agenda Builder >
Speakers & Team >
AI Script Studio >
Live Stage Control >

BACKSTAGE OPS
Backstage Control >

ANCHOR LECTERN
Stage Teleprompter >
Event Brief >

Joined as TechFest 2026
Landing / Switch Event

TechFest 2026 TF2026
Main Auditorium, Block A · 2026-02-14

Organizer Coordinator Anchor Stage

Live Sync 04:21:54 PM

STAGE VIEW Priya (Master of Ceremonies)
+365m Schedule Drift Stage Mode

NOW ON STAGE
Welcome Address
Dr. Rao (Principal & Patron) · Main Auditorium
TIME REMAINING
Scheduled: 16:15 → 16:25
03:06
DELAYED

UP NEXT
Event Concludes
Anchor / Organizer
Main Hall
Available Speaking Window: 187s

SMART STAGE TELEPROMPTER speaker introduction v22 (14s - template) Grounded in event data

"Our next session features Dr. Mehta, Chief Technology Officer at ABC Technologies, presenting on "Keynote — Future of AI". Please join me in giving a warm welcome to the stage."

Quick alternative: "Please put your hands together for Dr. Mehta (Chief Technology Officer at ABC Technologies) as we explore "Keynote — Future of AI"."

Pre-generated speaker introduction.

ONE-TAP QUICK ACTIONS (REAL-TIME SCRIPT ADAPTATION)

AI-generated script view



SMART_STAGE

LIVE OPERATING PLATFORM

ACTIVE CONSOLE

Organizer

TF2026

ORGANIZER TOOLS

Dashboard

Agenda Builder

Speakers & Team

AI Script Studio

Live Stage Control

BACKSTAGE OPS

Backstage Control

ANCHOR LECTERN

Stage Teleprompter

Event Brief

Joined as TechFest 2026

Landing / Switch Event



TechFest 2026

TF2026

Main Auditorium, Block A · 2026-02-14

Organizer

Coordinator

Anchor Stage

Live Sync

04:24:15 PM

Speakers, Guests & Jury

Profiles stored here serve as the single source of truth for all AI-generated stage scripts.

Speakers & Guests (3)

Jury Members (5)

All Profiles (8)

+ Add Speaker / Guest

JURY

Prof. Shah

HOD, Computer Engineering · Host Institute of Technology

Topic / Session:
"Hackathon Project Evaluation"

Distinguished academic with 20+ years of software architecture research.

Grounded Profile

✓ Verified

JURY

dhwani

computer science eng

No bio supplied — AI will ground intro strictly in credentials alone.

Grounded Profile

✓ Verified

JURY

Reha shah

computer Science

No bio supplied — AI will ground intro strictly in credentials alone.

Grounded Profile

✓ Verified

JURY

rishva

cse

No bio supplied — AI will ground intro strictly in credentials alone.

Grounded Profile

✓ Verified

JURY

astha

cse

No bio supplied — AI will ground intro strictly in credentials alone.

Grounded Profile

✓ Verified

Agenda & speaker management

ORGANIZER TOOLS

- Dashboard
- Agenda Builder
- Speakers & Team
- AI Script Studio
- Live Stage Control

BACKSTAGE OPS

- Backstage Control

ANCHOR LECTERN

- Stage Teleprompter
- Event Brief

oined as TechFest 2026

Landing / Switch Event

Live Stage Control

Organizer mirror and structural live overrides. Changes broadcast immediately to coordinator phones and anchor lecterns.

BROADCAST LIVE ANNOUNCEMENT

Sends an instant notification badge to the anchor teleprompter and coordinator feed.

Severity Level

General Info

Announcement Message

e.g. Parking on the north gate is now open, or lunch is served in foyer.

Broadcast to Stage & Staff

CURRENT LIVE STATUS

Live Stage Item: Welcome Address

Current Drift: +365 min

Speaking Window: 90s

REAL-TIME STAGE PROGRESSION

#1 Opening Ceremony COMPLETED
16:00 → 16:00 • No speaker • Main Auditorium

#2 Welcome Address DELAYED
16:15 → 16:25 • Dr. Rao • Main Auditorium

Start

#3 Keynote — Future of AI COMPLETED
12:42 → 12:43 • Dr. Mehta • Main Auditorium

#4 Workshop — Building with LLMs COMPLETED
12:57 → 12:57 • Ms. Patel • Room C

#5 Refreshment Break COMPLETED
15:53 → 15:53 • No speaker • Foyer

FULL UPDATE STREAM (APPEND-ONLY AUDIT)

Added new jury: astha (cse)

4:23:43 PM

Live Stage Control

SmartStage: Impact & Future Vision

IMPACT



Ready-to-read script for every moment on stage



Schedule changes handled in seconds, not panic



One shared live view for organizers and backstage



Fits hackathons, seminars, fests and conferences

Future Scope & Innovative Possibilities



New features

- Voice Assistance
- Smart Notifications
- Advanced Analytics
-



Scalability

- Multi-event, multi-tenant
- Cloud Deployment
- Campus-Wide Scaling



Integrations

- Google Calendar sync
- Slack / WhatsApp alerts



Security & speed

- Role-based authentication
- Caching & audit logs
- Data Encryption
- Faster API Calls

Explore Our Solution



GitHub Repository

<https://github.com/Dwan18/BIT-N-BUILD.git>

Brainstormers

- Astha Patel
- Dhvani Shah (Team Leader)
- Reha Shah
- Rishva Prajapati